PIB Project Synthesizer

Power amplifier

Pulse Soundwork commissioned by Windesheim University of Applied Sciences
Made by Jeroen Middel & Matthijs Vos

Introduction

In this project the power supply and power amplifier for a modular synthesizer were made. The synthesizer is divided into different modules and across different groups of students within the electrical engineering course. Together this makes a working synthesizer for the project ingenieursbureau.

To make a Power amplifier, a couple research questions were made:

How do you design a (semi-) modular Power amplifier and Power supply for a Synthesizer, with symmetrical and asymmetrical Voltages, and delivers clean audio?

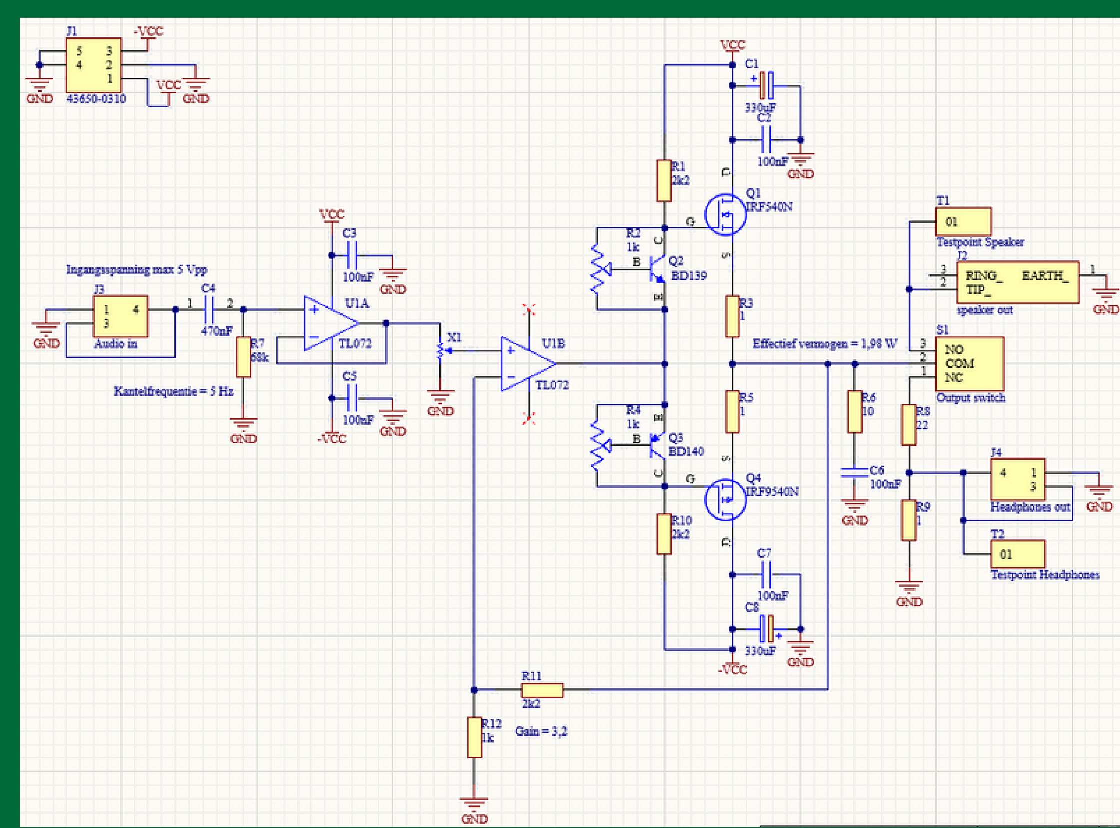
- What kind of power amplifier is the most suitable for audio?
- What kind of current protection is needed?
- How do you design a symmetrical power supply and a asymmetrical power supply that can deliver multiple voltages, using an outlet?

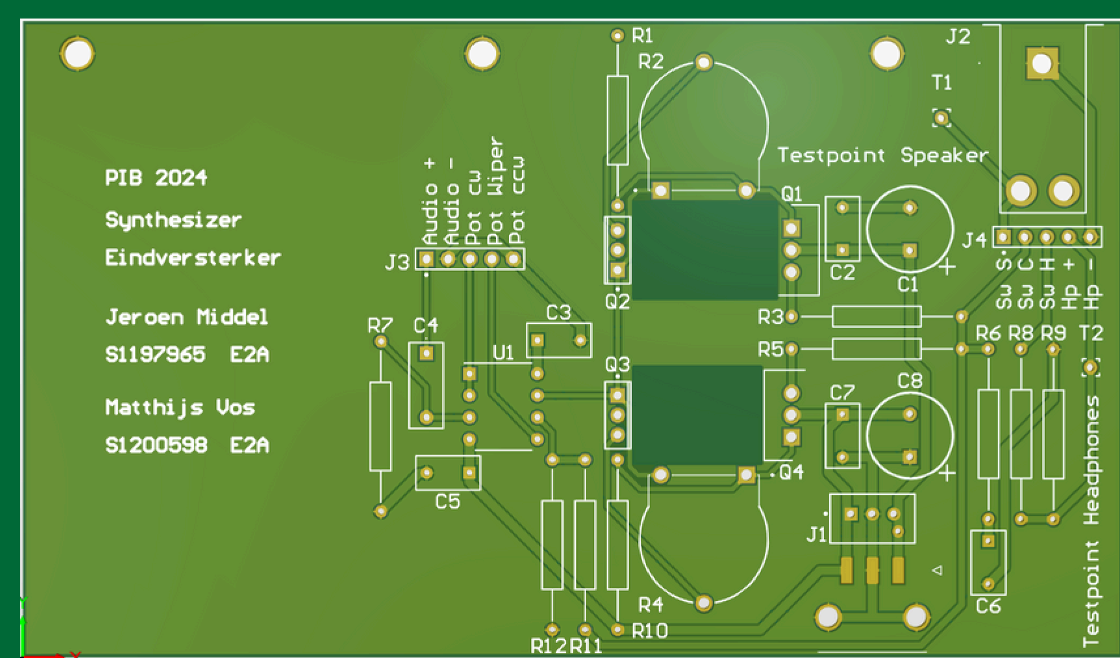
REQUIREMENT FOR THE POWER AMPLIFIER

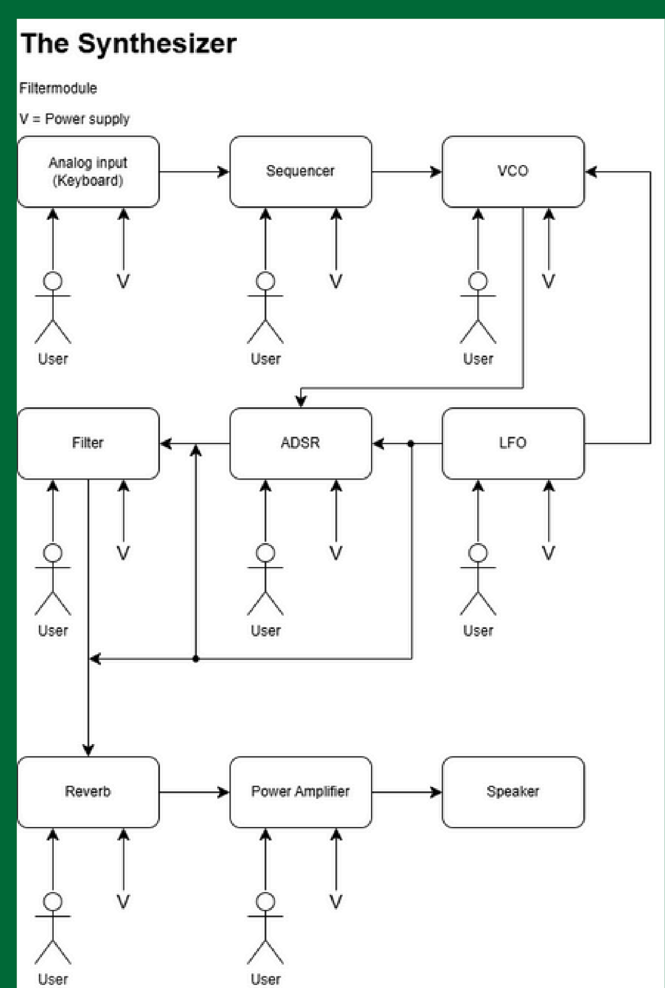
- The power amplifier must be able to amplify enough for an external speaker of 8 ohm.
- The power amplifier must be able to amplify enough for an external speaker of 8 ohm.
- The modules in front of the amplifier must be protected.
- The user must be able to adjust the volume.
- The amplifier must not become too hot. (passive, active)
- Maximum distortion. (1% harmonic)

Design

To safely use this module and the speakers, it is essential to filter out the DC component. This is done by the High-pass filter at the input of the module. After that, the signal needs to be buffered, before it is used to control to volume with a potentiometer. After the potentiometer comes the Power Amplifier, with class-AB the amplifier usually exists of 3 stages: the Voltage Amplifier, the Driver and the Output stage. After the output stage there is a resistor in series with a capacitor for a more stable signal. After that is a switch to control the output of the power amplifier: Speaker or Headphones. This module has a total gain of around 3,2; and is powered with the unstable rectified power from the power supply.

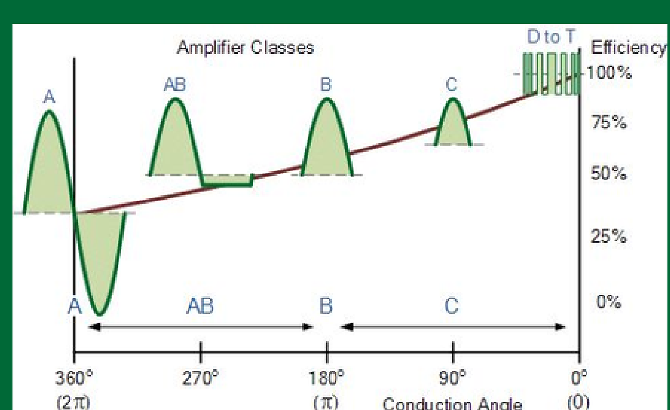






Research

In this project, a class AB audio amplifier was chosen. This decision was made after weighing the pros and cons of the most common audio amplifier classes. Class AB offers a good balance of efficiency, sound quality, and complexity. Therefore, this class of audio amplifier fits very well into this project.



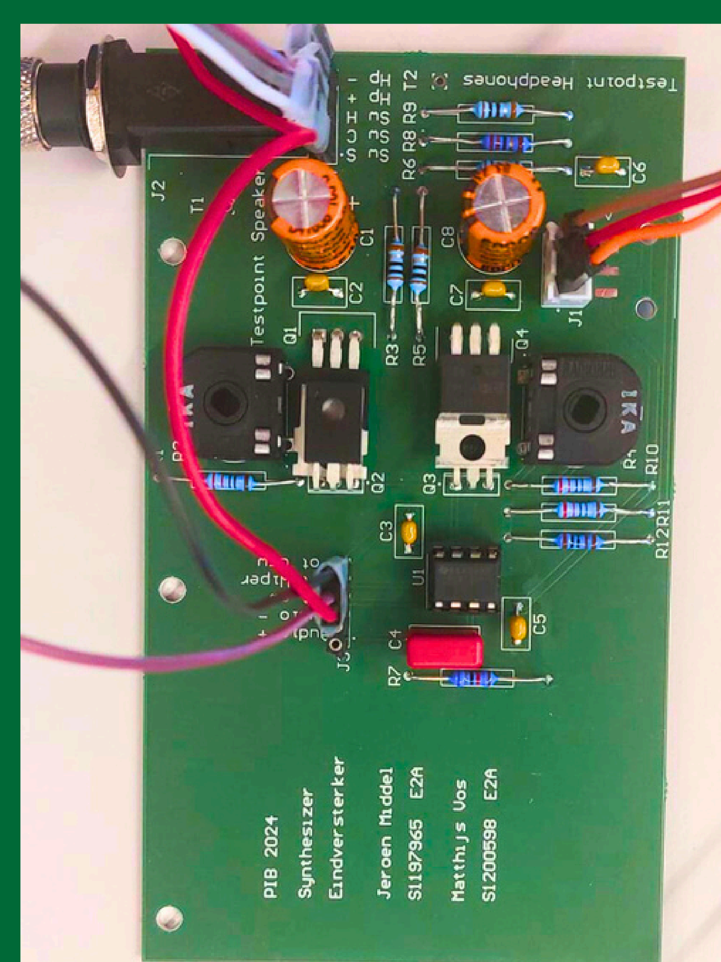
AMPLIFIER CLASS	EFFICIENCY	SOUND QUALITY	DISTORTION	COMPLEXITY
Class A	Low (20%-30%)	Very High	Very Low	Low
Class B	Medium (60%-80%)	Medium	High	Medium
Class AB	Medium (40%-60%)	High	Low	Medium to High
Class D	Very High (90%-95%)	Low/Very High*	Medium/Low*	Low/Very High*

*The sound quality, distortion, and complexity depend greatly on what kind of class D amplifier you make, whether it is a simple old-school one or a very complex modern one.

Conclusion

From the PIB project, an electronic article, a poster, and a functional PCB have been developed. The PCB meets all specified requirements. The following questions were also addressed in the project:

What kind of power amplifier is the most suitable for audio?
- Our research has shown that Class AB is the most suitable type for the synthesizer.
What type of protection is required in the Power amplifier?
- A buffer has been added to the amplifier's input, and a DC filter has also been included.
How do you ensure the Power amplifier is semi/fully modular, and which option is best for our application?
- Both fully modular and semi-modular Power amplifiers have pros and cons. For this project, a semi-modular Power amplifier was chosen because a fully modular Power amplifier quickly becomes too complex.



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Big thanks to the other groups with whom the synthesizer was made.
- Ingmar Kosmeije and Hidde Huysman: ADSR and Mixer
- Rick Smelt and Patrick van Ommen: Analog and Digital Input
- Jens Kuizenga and Diémo Rijsdijk: Filter
- Roelof Zijda and Reinout Strating: Sequencer
- Thijn Berkhof and Ralph Hänninger: VCO
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Power supply

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Introduction

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To make a Power supply, a couple research questions were made:

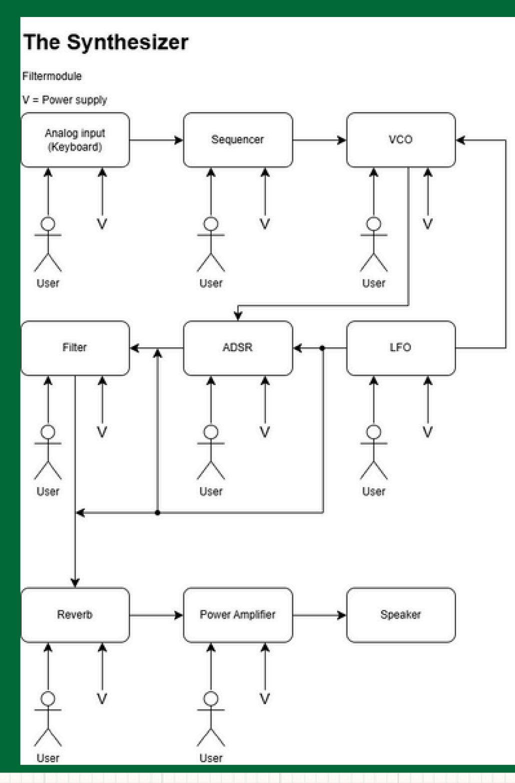
How do you design a (semi-) modular Power amplifier and Power supply for a Synthesizer, with symmetrical and asymmetrical Voltages, and delivers clean audio?

- What kind of power amplifier is the most suitable for audio?
- What kind of current protection is needed?
- How do you design a symmetrical power supply and a asymmetrical power supply that can deliver multiple voltages, using an outlet?

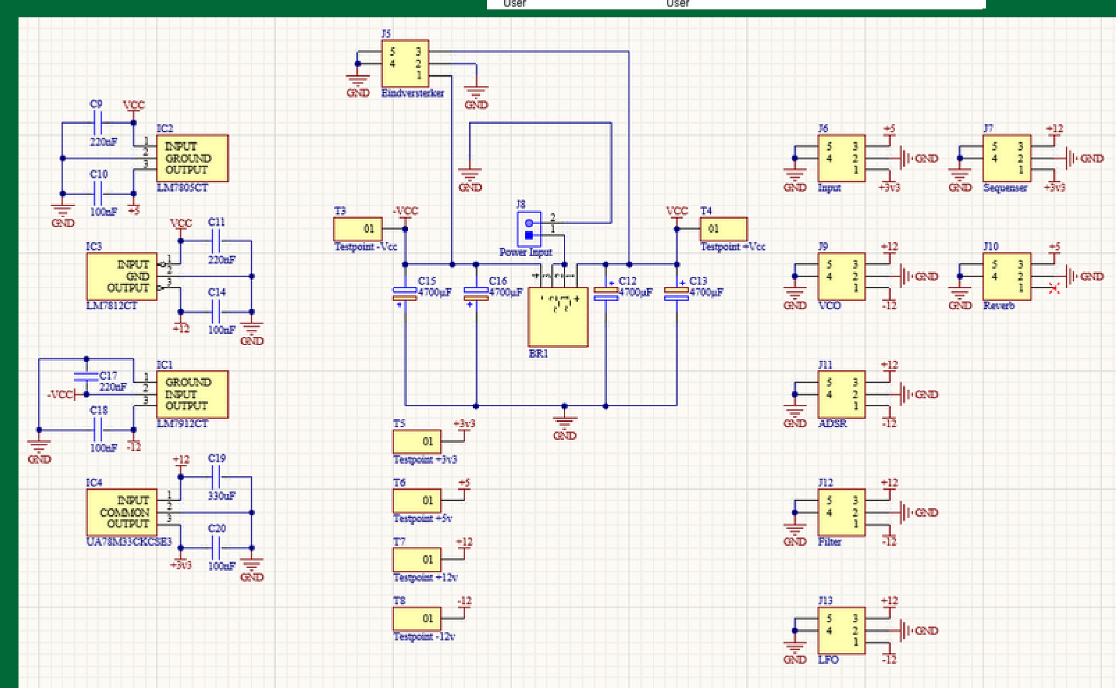
REQUIREMENT FOR THE POWER SUPPLY

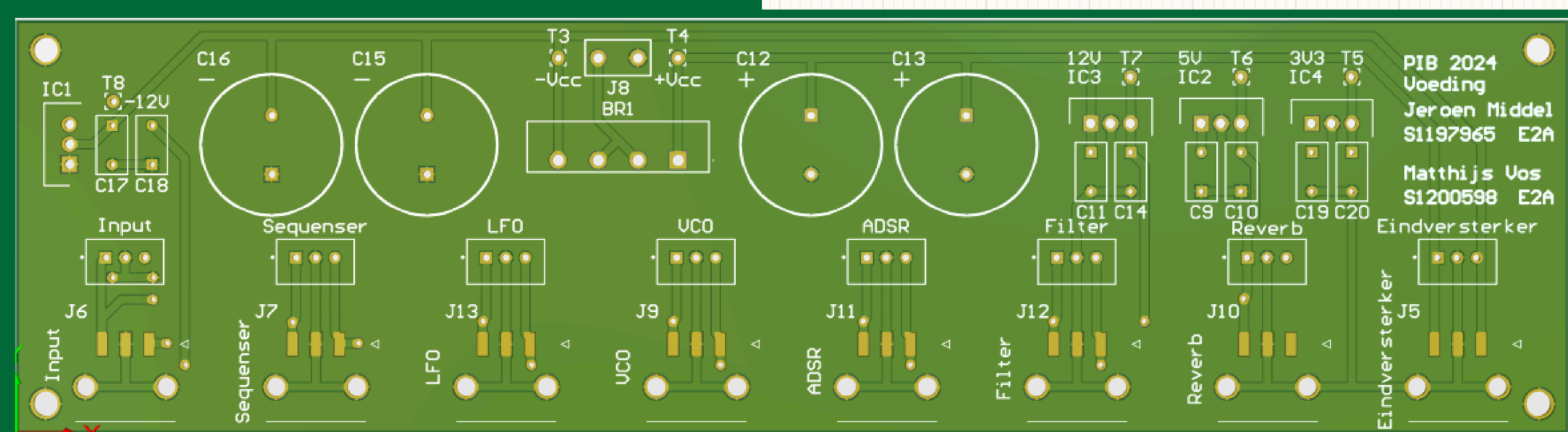
- Must be able to supply different voltages to different modules in the synthesizer.
- Must be able to supply symmetrical power from at least -10 to +10 Volts.
- Must be able to supply an asymmetrical power supply of at least +3.3 Volts.
- The power supply must be able to supply at least 80 mAmperes to each of the modules.
- The power supply must be able to be switched on and off by the user.
- The power supply must not become too hot. (passive, active)
- Must have low ripple voltage. (< 50mV)

Design



To power the synthesizer from an outlet, a separate power supply is needed, that makes a symmetrical voltage and an asymmetrical voltage. To provide these 3 lines are needed: Positive voltage, Ground and Negative voltage. To make the power from the outlet safe to use, a transformer that transforms 230 VAC to 12 VAC is used. To still have a groundline in the system, one of the two output terminals is the groundline, and is connected to the Earth terminal. To still make a positive and a negative voltage from one terminal, there are 2 half-wave rectifiers with a couple of electrolytic capacitors to minimise the ripple voltages. Because the capacitor pull a lot of current when the power supply is switched on, there are 2 diodes in parallel used per rectifier. The powerlines after the capacitor are connected to the Power Amplifier. All the other modules need a more stable voltage, therefore aa couple of voltage regulator is used to make positive 3,3 V, 5 V and 12 V and negative 12 V.

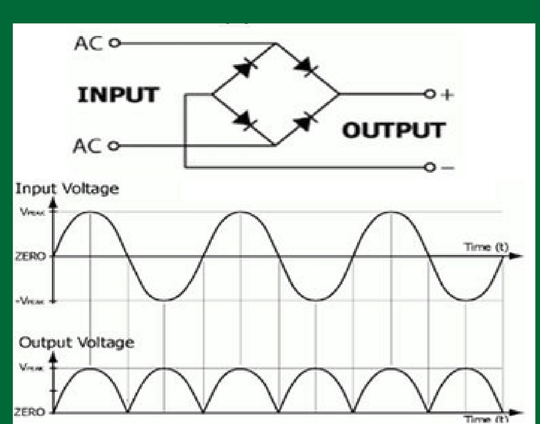


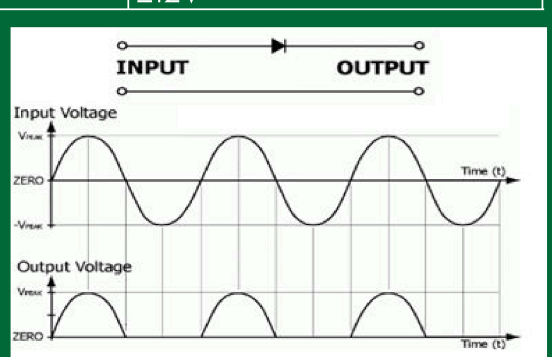


Research

The power supply is expected to provide both symmetric and asymmetric voltages to the various modules in the synthesizer. This has been agreed upon in collaboration with all groups and is shown in the table below.

MODULES	MINIMUM REQUIREMENTS
ADSR and Mixer	±12V
Analog and Digital input	+3,3V and +5V
Filter	±10V
Sequencer	+10V and +3,3V
VCO	±12V
LFO	±12V





Conclusion

From the PIB project, an electronic article, a poster, and a functional PCB have been developed. The PCB meets all specified requirements. The following questions were also addressed in the project:

What type of protection is required in the power supply?

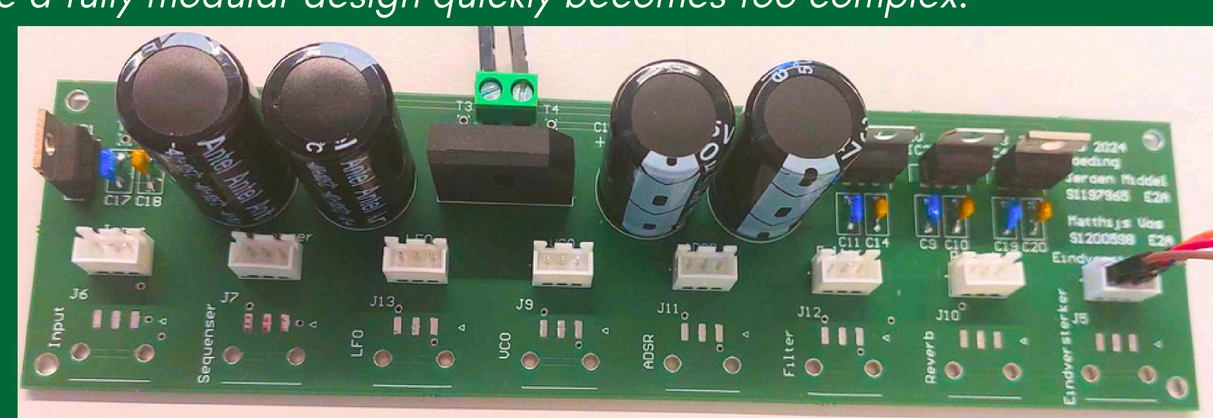
– The transformer includes a fuse that blows if too much current flows.

How do you design/create a power supply for a synthesizer that can deliver both symmetrical and asymmetrical power at different voltages, using a outlet as the source?

– In this project, two half-wave rectifiers were created—one for +12V and one for -12V. For the asymmetrical power supply, two voltage regulators were used: one for +5V and one for 3.3V.

How do you ensure the power supply is semi/fully modular, and which option is best for our application?

– Both fully modular and semi-modular power supplies have their pros and cons. For this project, a semi-modular power supply was chosen because a fully modular design quickly becomes too complex.



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